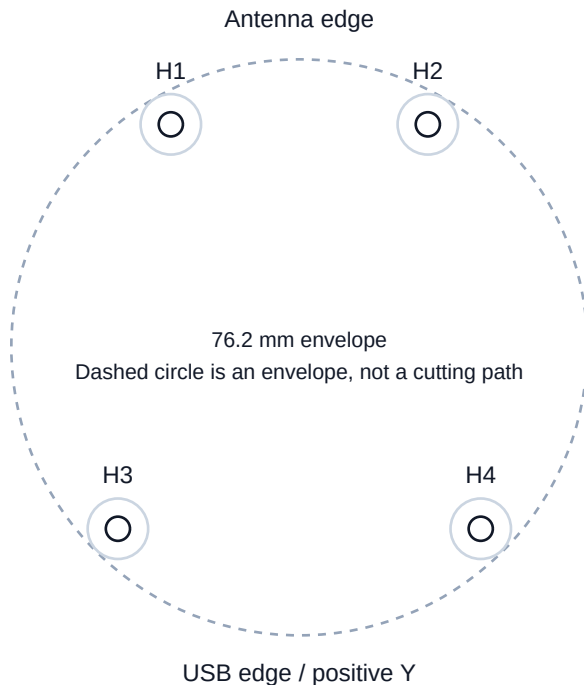


LIGHT v0.1 / mounting and orientation

One 3-inch envelope PCB | eight copper layers | two assembled prototypes

Top/front faces the PSU above. The two motor ports, three spotlights and seventh LED zone mate perpendicular to the bottom/back face. Height remains flexible; connector bodies may overhang the PCB envelope.



Four M3 clearance holes: 3.2 mm NPTH.

8 mm head/standoff reserve on both faces.

Upper spacing: 34 mm. Lower: 48 mm. Row spacing: 53.5 mm.

Hole centers relative to board center; X right, Y toward USB:

H1 (-17, -29.5) mm; H2 (+17, -29.5) mm.

H3 (-24, +24) mm; H4 (+24, +24) mm.

Use nonmetallic hardware near the antenna.

Finish: BLACK solder mask, WHITE silkscreen on both faces. Branding: ENGINEERED LIGHTING / LIGHT v0.1. The following assembly views show native geometry and labels; their plotting colours do not represent the ordered finish.

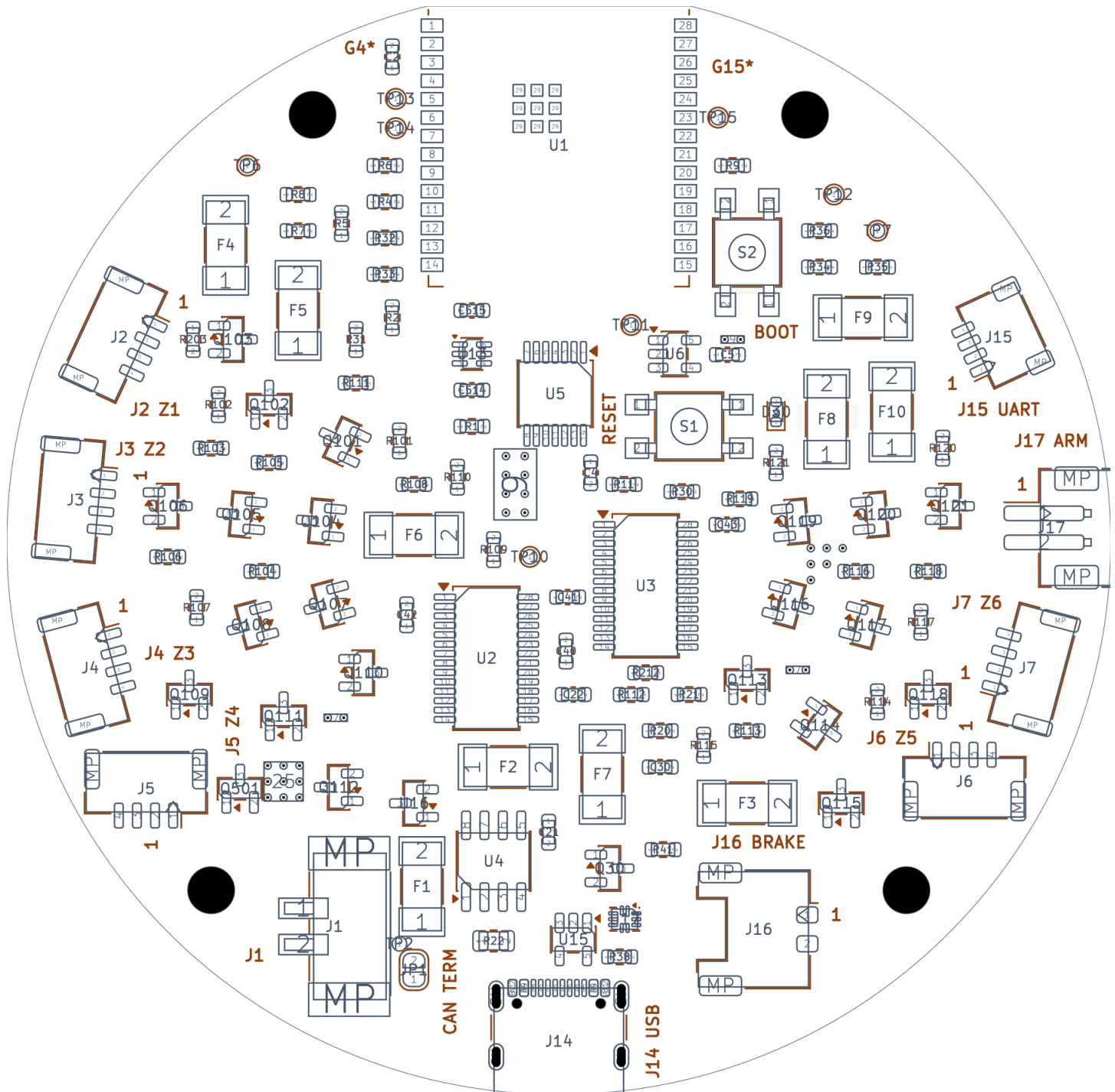
Validation: zero ERC/DRC violations and zero unrouted items; exact PCB/schematic/BOM pin parity; all 163 assigned nets connected; four mounting envelopes clear on both faces; no unexpected drilled-hole/solder-pad intersections. USB data copper is preserved and its sampled ground reference remains covered.

Temperature: TMP112 at I2C address 0x48 measures local PCB temperature. Expansion: J18 exposes five spare GPIOs plus 3V3 and GND. Boot straps are separate probes. USB-C, RESET and BOOT remain available.

Factory stackup, sourcing, assembly processes, enclosure/cable access and protected powered tests remain required. Cable weight must be supported separately. A separate 24 V / 5 A PSU remains provisional under the existing 4 A commissioning envelope.

TOP / assembly view

One 3-inch envelope PCB | eight copper layers | two assembled prototypes

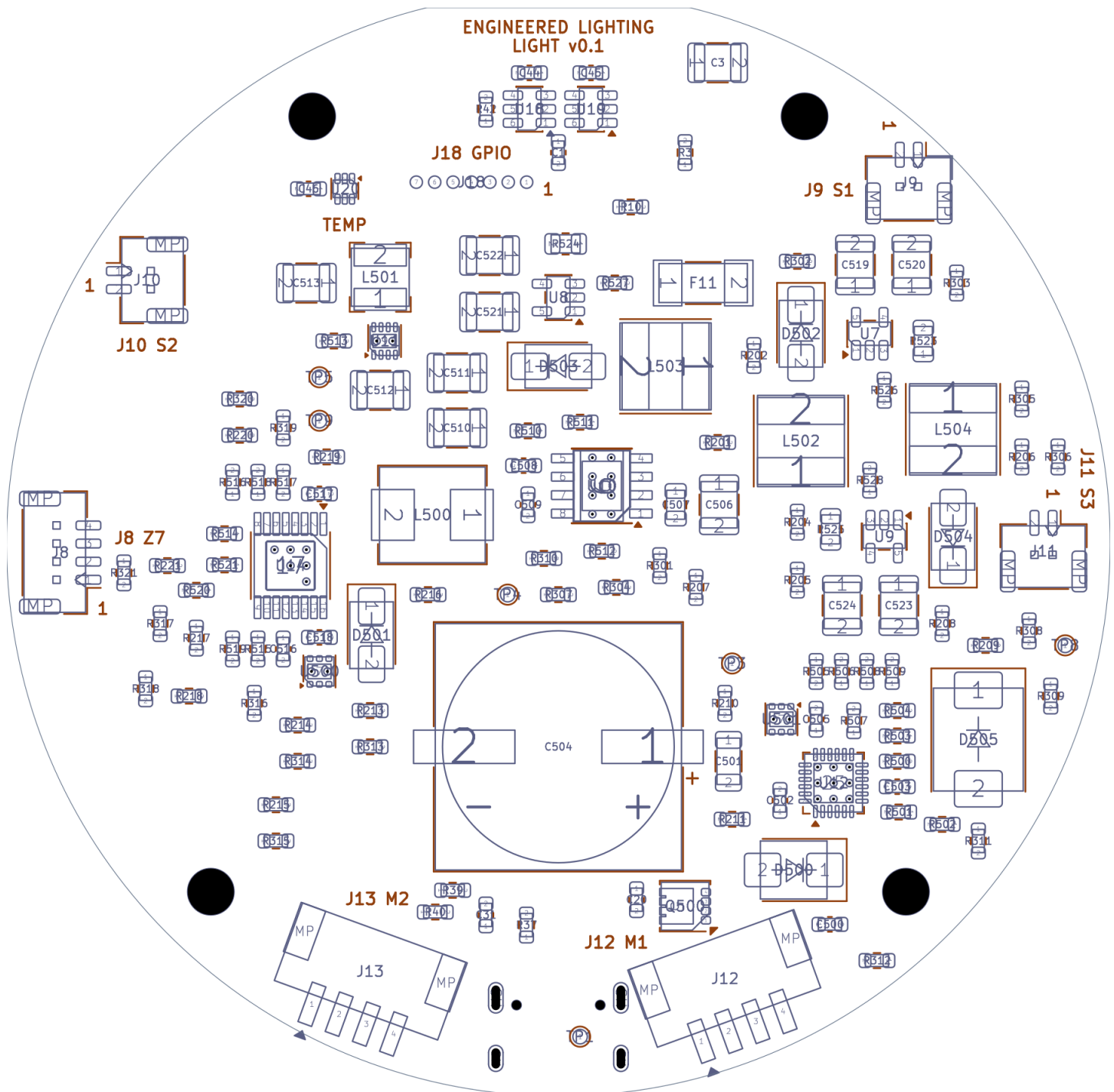


View from above the physical top/front face.

Use numeric PCB pad/contact numbers for harnesses. Do not infer cable cavity order from the image. Faint component outlines are assembly references; coloured service text is native silkscreen. Not to scale.

BOTTOM / assembly view

One 3-inch envelope PCB | eight copper layers | two assembled prototypes



View from below the physical bottom/back face; the image is mirrored from native CAD coordinates.

Use numeric PCB pad/contact numbers for harnesses. Do not infer cable cavity order from the image. Faint component outlines are assembly references; coloured service text is native silkscreen. Not to scale.

Ports, temperature and expansion

One 3-inch envelope PCB | eight copper layers | two assembled prototypes

Port	Face / mating	Function	PCB contact mapping
J1	Top / up	PSU	1 +24V; 2 GND
J2-J7	Top / radial	Zones 1-6	1 +24V; 2 W; 3 N; 4 C
J8	Bottom / down	Zone 7	1 +24V; 2 W; 3 N; 4 C
J9-J11	Bottom / down	Spots 1-3	1 LED+; 2 LED-
J12-J13	Bottom / down	Motors 1-2	1 +24V; 2 GND; 3 CAN_H; 4 CAN_L
J14	Top / edge	USB-C	Data/service; external 24V required
J15	Top / edge	UART	1 3V3; 2 GND; 3 TX; 4 RX
J16	Top / edge	Bus / brake	1 protected 24V; 2 GND
J17	Top / up	ARM	1 3V3; 2 ARM; open = disarmed
J18	Bottom / pads	Expansion	1 3V3; 2 GND; 3-7 GPIO19-23

PSU input uses Phoenix 1771091, upward spring wire entry. Recommended starting harness: 0.5 mm² / AWG20, 6 mm strip. Confirm polarity by numbered terminal contacts before power.

Motor sockets use vertical JST PA; spot and zone 7 sockets use vertical JST GH. Their downward mating axes keep cable routing on the underside.

Do not ground W/N/C outputs or spotlight LED-. J16 is an exposed protected bus; any brake accessory requires its own controlled chopper.

TMP112: I2C 0x48 on existing SDA/SCL, 3.3 V. Read normal-mode signed 12-bit temperature at 0.0625 C/LSB. No firmware was implemented; see the addendum.

J18 pads: 0.85 mm diameter, 1.27 mm pitch, no installed header. GPIO19-23 are 3.3 V signals. No new peripheral power budget or external-I/O protection is implied.

TP13/TP14/TP15 expose GPIO4/5/15 boot straps. G4* and G15* identify the labelled probes; use the native assembly references for TP14. External reset-time levels can alter boot mode.